

$$(1) \sin \theta \cos \theta = \frac{1-t^2}{2} \quad (2) t = \sqrt{2} \sin \left(\theta - \frac{\pi}{4} \right), \quad -\sqrt{2} \leq t \leq \sqrt{2} \quad (3) 1 \leq y \leq 4 + 2\sqrt{2}, \quad 1 < a < 4 - 2\sqrt{2}$$

$$(1) \cos \alpha = \frac{\sqrt{6}}{3}, \sin 2\alpha = -\frac{2\sqrt{2}}{3} \quad (2) y = 2 \sin \theta + 2\sqrt{2} \cos \theta \quad (3) (\text{最大值}) 2\sqrt{3}, (\text{最小值}) -2\sqrt{2}$$

$$(1)(\mathcal{A})_{\frac{1}{2}}, \quad (\mathcal{I})-2, \quad (\mathcal{U})0, \frac{5\pi}{6}, \pi \qquad (2)b=1, \quad \theta=0, \pi, \frac{4\pi}{3}$$

$$(1)8, \ 4 \qquad (2)2k, \ k \qquad (3)32, \ \frac{993}{2}$$